Solution Requirements

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Emotion Detection and Learning Support Engine requires a set of functional and non-functional requirements to ensure the system accurately detects emotions, provides personalized learning guidance, stores prediction history, and offers a reliable and user-friendly experience.

Functional Requirements

SN	Requirement Category	Requirement Description	Priority
1	Authentication	Users should be able to access the application through the Streamlit interface without a complex login process (or login if implemented).	High
2	Authorization Levels	The application provides standard user-level access for emotion prediction and guidance.	Medium
3	External Interfaces	The system accepts user text input through the Streamlit web interface and displays predictions, guidance, and analytics.	High
4	Transaction Processing	The system processes user input, predicts emotions, calculates confidence scores, and generates learning guidance in real time.	High
5	Reporting	The system records prediction history, timestamps, confidence scores, and displays them through the analytics dashboard.	High
6	Business Rules	The application classifies emotions using the trained BiLSTM model and generates learning guidance based on the detected emotion.	High
7	Compliance to Laws / Regulations	User information is processed only for emotion analysis, and prediction logs are maintained locally without exposing sensitive information.	Medium
8	Other	The application provides primary emotion, secondary emotion, confidence score, and personalized learning recommendations.	Medium

Non-Functional Requirements

SN	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance and Speed	Emotion prediction should be generated quickly after user input.	Prediction generated within 2–3 seconds.
2	Scalability	The application should support future enhancements such as larger datasets and additional emotion classes.	Easily extensible architecture.
3	Security and Data Privacy	User input and prediction logs should be stored securely without exposing sensitive data.	Local storage and secure handling of user data.
4	Reliability and Availability	The application should consistently produce emotion predictions and learning guidance without failures.	Stable operation during normal usage.
5	Usability and Accessibility	The interface should be simple, intuitive, and easy for students to use.	User-friendly Streamlit interface with minimal learning curve.
6	Other	The solution should be maintainable and allow future integration of advanced AI models and cloud deployment.	Modular codebase supporting future improvements.